

office hours

9/25/25

What should we talk about??

Q1:

$$H_0: y_i = \beta_0 + \varepsilon_i$$

1 vector

$$H_1: y_i = \beta_0 + \beta_1 x_{1i} + \dots + \beta_{p-1} x_{p-1i} + \varepsilon_i$$

p vectors

$$F = \frac{MSR}{MSE}$$

$\Leftrightarrow$

$$F =$$

$$\frac{(SSE_{H_0} - SSE_{H_1}) / (df_{SSE_{H_0}} - df_{SSE_{H_1}})}{SSE_{H_1} / df_{SSE_{H_1}}}$$

$$SSE_{H_0} = \sum_{i=1}^n (e_i)^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - \bar{y})^2$$

What is $\hat{y}_i$ under H_0 ?

$$= \underline{\underline{SST}}$$

$$\hat{y}_i = \hat{\beta}_0 = \bar{y}$$

$$df_{SSE_{H_0}} = n - p$$

$$= n - 1 \quad \text{since } p=1 \text{ under } H_0$$

$$\begin{aligned}
 \underline{SSE}_{H_1} &= \sum_i e_i^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2 \\
 &= \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{1i} - \dots - \hat{\beta}_{p-1} x_{(p-1)i})^2 \\
 &= SST - \underline{SSR}_F
 \end{aligned}$$

$$df_{SSE_{H_1}} = n - p \quad \text{under } H_1$$

Num:

$$\frac{SSE_{H_0} - SSE_{H_1}}{(n-1) - (n-p)} = \frac{\cancel{SST} - (\cancel{SST} - SSR)}{p-1}$$

$$= \underline{SSR / p-1}$$

$$= MSR$$

$$\underline{\frac{SSR}{\sigma^2}} \stackrel{H_0}{\sim} \chi^2_{p-1}$$

$$\underline{Den}: \frac{SSE_{H_1}}{n-p} = \frac{SSE_F}{n-p} = MSE$$

$$\underline{\frac{SSE}{\sigma^2}} \sim \chi^2_{n-p}$$

$$Q(\beta_0) = \sum_{i=1}^n (y_i - \beta_0)^2$$

$$\frac{\partial Q}{\partial \beta_0} = -2 \sum_{i=1}^n (y_i - \beta_0) \stackrel{!}{=} 0$$

$$\Leftrightarrow \sum_{i=1}^n y_i - n\beta_0 = 0$$

$$\Leftrightarrow \beta_0 = \sum y_i / n = \bar{y}$$

$$\tilde{Q}(\beta_0) = \sum_{i=1}^n |y_i - \beta_0|$$

$$\operatorname{argmin}_{\beta_0} \tilde{Q}(\beta_0) = \operatorname{median}(y)$$

What is SSR in Simple LR?

$$\frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sigma^2} = \frac{\hat{\beta}_1^2 \sum_{i=1}^n (x_i - \bar{x})^2}{\sigma^2}$$

$$= \left(\frac{\hat{\beta}_1 - 0}{\sigma / \sqrt{SSX}} \right)^2$$

$$\hat{\beta}_1 \stackrel{H_0}{\sim} N(0, \sigma^2 / SSX)$$

$$\frac{\hat{\beta}_1}{\sigma / \sqrt{SSX}} \sim N(0, 1)$$

$$= (z)^2 = \chi^2_{1 = p-1}$$

(p=2)

MUR What is SSR?

$$SSR = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 = \left\| \hat{\underline{y}} - \bar{y} \right\|^2 = \left\| H\underline{y} - A\underline{y} \right\|^2$$

$\begin{pmatrix} \bar{y} \\ \hat{\underline{y}} \end{pmatrix}$

$$A\underline{y} = \frac{1}{n} \sum_{i=1}^n \underline{z}_i y_i = \frac{1}{n} \underline{I}_n (\underline{I}_n^T \underline{y}) = \underbrace{\frac{1}{n} \underline{I}_n \underline{I}_n^T}_A \underline{y}$$

$$\underline{I}_n = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}_n$$

$$= \|H\mathbf{y} - A\mathbf{y}\|^2 = \| \underline{(H-A)}\mathbf{y} \|^2 \rightarrow [(H-A)\mathbf{y}]^T [(H-A)\mathbf{y}]$$

$$\| \mathbf{x}(\mathbf{x}^T \mathbf{x})^{-1} \mathbf{x}^T \|$$

$$= \mathbf{y}^T (H-A)^T (H-A) \mathbf{y} = \mathbf{y}^T \underline{(H-A)} \mathbf{y}$$

$$\frac{\mathbf{y}^T (H-A) \mathbf{y}}{\sigma^2} \sim \chi^2_{\text{tr}(H-A)}$$

$H-A$ is sym:

$$(H-A)^T = H^T - A^T = H - \left(\frac{1}{n} \mathbf{I}_n \mathbf{A}_n^T \right)^T$$

$$= H - \frac{1}{n} (\mathbf{A}_n^T)^T \mathbf{I}_n^T = H - A$$

$H-A$ is idempotent:

$$(H-A)^2 = (H-A)(H-A)$$

$$= H^2 - AH - HA + A^2$$

$$= H^2 - \frac{1}{n} \mathbf{A}_n \underline{\mathbf{A}_n^T} H - H \underline{\frac{1}{n} \mathbf{A}_n \mathbf{A}_n^T} + \left(\frac{1}{n} \mathbf{A}_n \mathbf{A}_n^T \right) \left(\frac{1}{n} \mathbf{A}_n \mathbf{A}_n^T \right)$$

$$= H - \frac{1}{n} \mathbf{A}_n \mathbf{A}_n^T - \frac{1}{n} \mathbf{A}_n \mathbf{A}_n^T + \frac{1}{n^2} \mathbf{A}_n \mathbf{A}_n^T \mathbf{A}_n \mathbf{A}_n^T$$

$$= H - A - A + \frac{1}{n} \mathbf{A}_n \mathbf{A}_n^T$$

$$= H - A - A + A = H - A. \quad \square \quad \cup$$

$$\underline{\underline{H^T H}} = (H^T I_n)^T = (\underline{\underline{H I_n}})^T = \underline{\underline{I_n^T}}$$

$$H I_n = I_n$$

$H I_n$ is the proj of I_n into $\text{Colspace}(X)$

$$X = [I_n \ x_1 \ x_2 \ \dots \ x_{p-1}] \text{ so } I_n \subset \text{Colspace}(X)$$

$$\text{tr}(H - A) = \text{tr}(\underline{\underline{H}}) - \text{tr}(A)$$

$$= \text{tr}(\underline{\underline{X(X^T X)^T X^T}}) - \text{tr}(\frac{1}{n} I_n I_n^T)$$

$$= \text{tr}(\underline{\underline{(X^T X)^T X^T X}}) - \text{tr}\left(\begin{bmatrix} \frac{1}{n} & \frac{1}{n} & \frac{1}{n} & \dots & \frac{1}{n} \\ \vdots & & & & \\ \frac{1}{n} & \dots & \dots & \dots & \frac{1}{n} \end{bmatrix}\right)$$

$$= \text{tr}(I_p) - (\frac{1}{n})n$$

$$= \textcircled{p} - 1 \in \mathbb{R}$$

$$\text{tr}(ABC) = \text{tr}(BCA) = \text{tr}(CAB) = \text{tr}(ACB)$$

$$\hat{y} = X \hat{\beta} = X \underbrace{(X^T X)^T X^T}_{\text{H}} y$$

$$AY = \begin{bmatrix} \frac{1}{n} & \dots & \frac{1}{n} \\ \frac{1}{n} & \dots & \frac{1}{n} \\ \vdots & & \vdots \\ \frac{1}{n} & \dots & \frac{1}{n} \end{bmatrix} \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$$

$$= \begin{pmatrix} \frac{1}{n}y_1 + \frac{1}{n}y_2 + \dots + \frac{1}{n}y_n \\ \vdots \\ \frac{1}{n}y_1 + \frac{1}{n}y_2 + \dots + \frac{1}{n}y_n \end{pmatrix} = \begin{pmatrix} \bar{y} \\ \bar{y} \\ \vdots \\ \bar{y} \end{pmatrix}$$

$$z \sim N(0, I_n)$$

$$\begin{aligned} \|z\|^2 &= z^T z \sim \chi_n^2 \\ &= \sum_{i=1}^n z_i^2 \end{aligned}$$

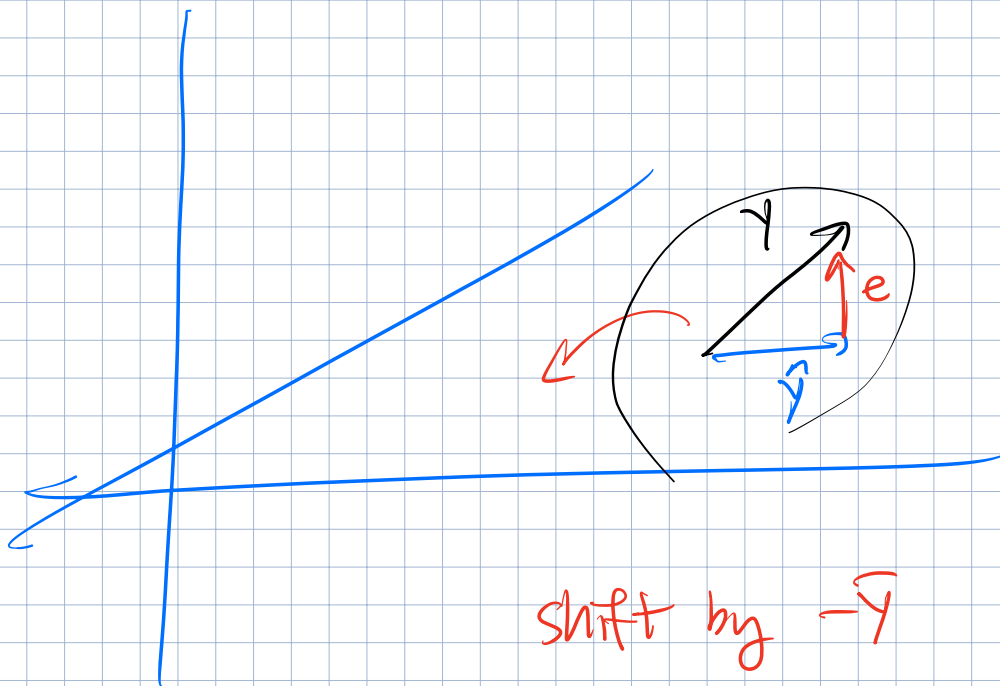
$$\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} y \sim N \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} I_{p-1} & 0 \\ 0 & 0 \end{pmatrix} \right)$$

$$\frac{y^T (H-A) y}{\sigma^2} = \left\| \frac{(H-A)^{1/2} y}{\sigma} \right\|^2$$

$$= \underbrace{\left(\frac{u_1^T y}{\sigma} \right)^2 + \dots + \left(\frac{u_{p-1}^T y}{\sigma} \right)^2}_{\text{sum of squares}} + 0^2 + \dots + 0^2$$

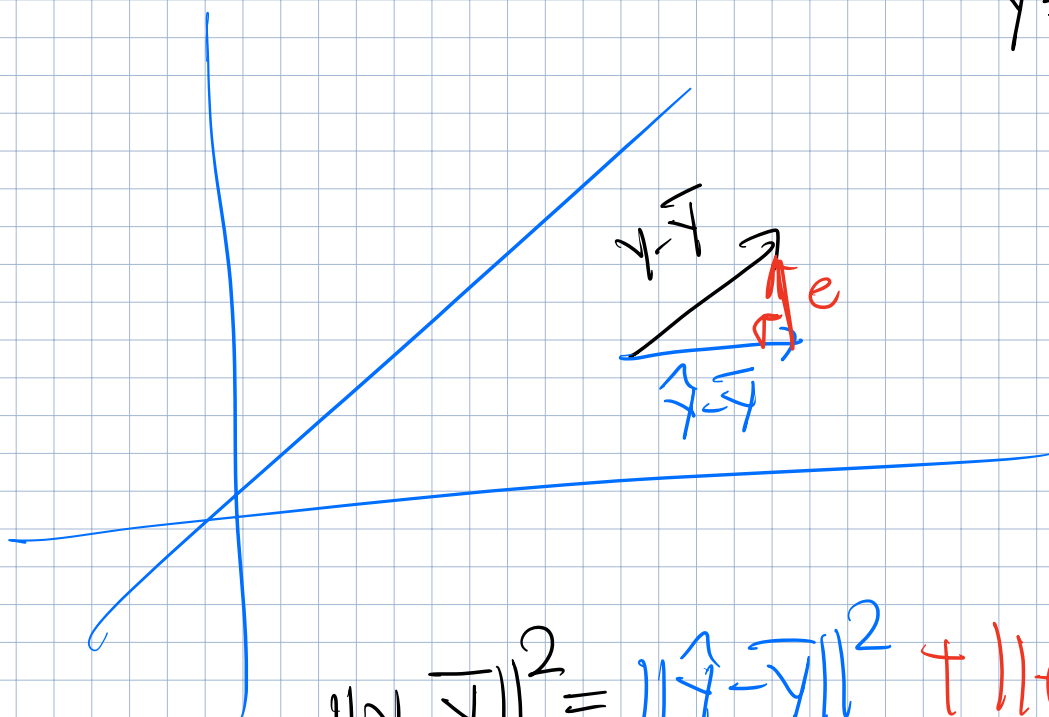
$$= \chi_{p-1}^2$$

$$SST = \sum_{i=1}^n (y_i - \bar{y})^2$$



$$\bar{y} = \frac{1}{n} \mathbf{1}_n \mathbf{1}_n^T \mathbf{y} = \bar{y} \mathbf{1}_n$$

$\subset \text{colspace}(X)$



$$\|y - \bar{y}\|^2 = \|\hat{y} - \bar{y}\|^2 + \|e\|^2$$

$$SST = SSR + SSE$$

Q3

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 \mathbb{1}(x=c_2) + \hat{\beta}_2 \mathbb{1}(x=c_3)$$

X is cat w c1, c2, c3

coef	...	p
int		
$\hat{\beta}_1$		-0.001
$\hat{\beta}_2$		<u>9</u>

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 \mathbb{1}(x=c_2) + \hat{\beta}_2 \mathbb{1}(x=c_3)$$

$$X = \begin{bmatrix} 1 & 0 & 0 \\ \vdots & 1 & 0 \\ \vdots & 0 & 1 \end{bmatrix}$$